Francesco Bronzino

École Normale Supérieure de Lyon, 46 allée d'Italie 69364 Lyon, France
francesco.bronzino@ens-lyon.fr (+33) 0761063734 www.fbronzino.com

EDUCATION

École Normale Supérieure de Lyon 2024
Habilitation à Diriger des Recherches
Thesis title: "Bridging the Gap Between Machine Learning and Networked Systems"
Committee: I. Chrisment, M. Crovella, M. Dias de Amorim, A. Duda, M. Fiore, I. Guérin-Lassous

Rutgers University 2012 - 2016
Ph.D. in Electrical and Computer Engineering (GPA 3.9/4)
Thesis title: "Named-Object Based Services in the Future Internet Architecture"
Advisor: Prof. Dipankar Raychaudhuri

Università degli Studi di Torino 2009 - 2012
Laurea Magistrale in Computer Science (110/110 summa cum laude)
Advisors: Prof. Marco Grangetto and Prof. Giovanni Pau (UCLA)

Università degli Studi di Torino 2006 - 2009
Laurea in Computer Science (107/110)
Advisor: Prof. Matteo Sereno

RESEARCH TOPICS

Data driven network management. Design of systems and data driven models to advance the operationalization of machine learning for network management tasks.

Distributed inference and services. Design of systems for efficient serving of machine learning inference and distributed services, *e.g.*, video analytics on heterogeneous edge-cloud resources.

Privacy oriented network systems. Performance analysis and design of network systems that preserve user privacy, *e.g.*, multi-party relay architectures.

EXPERIENCE

École Normale Supérieure de Lyon 2022 - Now
Maître de conférences (Associate Professor). Computer Science department and LIP (UMR 5668)

Institut Universitaire de France (IUF) 2025 - Now
Junior Fellow.

NetMicroscope Inc. (University Spin-off) 2021 - 2025
Co-founder and CTO.

Université Savoie Mont Blanc 2020 - 2022
Maître de conférences (Associate Professor). IAE Annecy and LISTIC

Nokia Bell Labs France 2018 - 2020
Research Scientist. AAAID Lab.

Inria, Paris, MiMove Research Group 2016 - 2018
Post-Doctoral Research Fellow. Mentor: Dr. Renata Cruz Teixeira.

WINLAB, Rutgers University, ECE Department 2012 - 2016
Graduate Research Assistant. Advisor: Prof. Dipankar Raychaudhuri.

SELECTED PUBLICATIONS

SoK: Mapping the Privacy Landscape of Geolocation Ecosystems
A. Laouar, P. Lachat, L. Desgeorges, M. Cunche, V. Roca, P. Vicat-Blanc, F. Bronzino
In PETS 2026, Calgary, Canada.

CATO: End-to-end Optimization of ML Traffic Analysis Pipelines
G. Wan, S. Liu, F. Bronzino, N. Feamster, and Z. Durumeric.
In USENIX NSDI 2025, Philadelphia, USA.

NetDiffusion: Network Data Augmentation Through Protocol-Constrained Traffic Generation
X. Jiang, S. Liu, A. Gember-Jacobson, A. Bhagoji, P. Schmitt, F. Bronzino, and N. Feamster.

In ACM Sigmetrics 2024, Venezia, Italy.

LEAF: Navigating Concept Drift in Cellular Networks
S. Liu, F. Bronzino, P. Schmitt, A. Nitin Bhagoji, N. Feamster, H. Garcia Crespo, T. Coyle, and B. Ward.
In ACM CoNEXT 2023, Paris, France.

Traffic Refinery: Cost-Aware Data Representation for Machine Learning on Network Traffic
F. Bronzino, P. Schmitt, S.Ayoubi, H. Kim, R. Teixeira, N. Feamster.
In ACM Sigmetrics 2022, Mumbai, India.

Inferring Streaming Video Quality from Encrypted Traffic: Practical Models and Deployment Experience.
F. Bronzino, P. Schmitt, S.Ayoubi, G. Martins, R. Teixeira, N. Feamster.
In ACM Sigmetrics 2020, Boston, USA.

NOVN: A Named-Object Based Virtual Network Architecture to Support Advanced Mobile Edge Computing Services.
F. Bronzino, S. Maheshwari, D. Raychaudhuri, I. Seskar.
Elsevier Pervasive and Mobile Computing Journal 69 (2020) 101261.

FUNDING AS PRINCIPAL INVESTIGATOR

ANR GeN-T: Generative AI for Network Digital Twins
Project developing the foundations for systematic synthetic data generation in NDTs. With CNRS / Ecole Polytechnique, EURECOM, Orange, and Nokia Bell Labs. **183k€ (out of project total 984k€), 2027-2031**

Comcast Faculty Innovation Fund Award
Project developing a passive measurement platform for home network performance analysis. **50k$, 2026-2027**

IUF ORGANIC: Observing Application Quality at Network Scale
Project developing a new paradigm for taking advantage of modern quality inference models to allow operators to manage networks at-scale. **75k€ and 2/3 of teaching duties reduction, 2025-2030**

ANR GTTP: Global Tracking Transfer Protocol
Project developing an unified privacy-first geolocalization infrastructure. With Inria and Sentiens. **233k€ (out of project total 501k€), 2025-2028**

CNRS IEA GATEAU: Generative AI Techniques for Network Management
Project developing AI-driven generators for network traces. With University of Chicago . **5k€, 2025-2026**

ANR-NSF MINT: Modeling Modern Network Traffic: From Data Representation to Automated Machine Learning
Project studying new methods to represent and model traffic for machine learning driven network management and inference. With University of Chicago and Stanford University. **219k€ (out of project total 602k€), 2021-2025**

ANR PARFAIT: Planning and Learning for AI-Edge Computing
Project studying distributed allocation schemes based on Markov decision processes and Reinforcement Learning for edge computing infrastructures tailored for AI-intensive processing tasks. With Avignon Université, Inria, and CNAM. **135k€ (out of project total 559k€), 2022-2025**

FIL INTERFERE: Stressing Systems Security Through On-the-fly Network Traffic Generation
Project studying the generation of real-time network traffic to create auditing tools for stressing systems and their security. With INSA Lyon **10k€, 2023-2025**

AAP USMB: Decentralized Live Video Analytics at the Network Edge
Project studying how to integrate mobile video sources into the video analytics processing infrastructure using heterogeneous compute resources at the edge of the network. **3k€, 2022**

FUNDING AS PROJECT MEMBER

ANR NEMIOT: Planning and Learning for AI-Edge Computing
Project studying the network behavior of IoT devices. With Inria and École Polytechnique. **172k€ (out of project total 372k€), 2024-2027**

Four grants through France And Chicago Collaborating in The Sciences (FACCTS)
With the University of Chicago. Four separate grants of **$22.5k, $15k, $15k, and $15k, 2021-2024**

Industrial contract with CELESTE

Project studying how to automatically extract network configurations satisfying client constraints using machine learning solutions that learn from past configurations rules and apply them to future cases. **Half a PhD thesis funding, 84k€, 2021-2024**

TEACHING

École Normale Supérieure de Lyon 2022 - Now
Networks, Network measurements, Performance Evaluation, and Architectures and Operating Systems

IAE, Université Savoie Mont Blanc 2020 - 2022
Intro to programming, Machine learning, Computer networks, and Operating systems

WINLAB, Rutgers University, ECE Department 2013 - 2016
Communication Networks II and Introduction to Computers for Engineers

ADVISING

Ph.D. Students

- Soumaya Aloui 2026 - Now
  "Distributed Observation of SDN Controllers for Anomaly Detection". With L. Desgeorges and I. Guérin Lassous (U Lyon).
- Augustin Laouar 2025 - Now
  "Global Tracking Transfer Protocol". With L. Desgeorges (U Lyon).
- Johann Hugon 2022 - 2025
  "Modeling Modern Network Traffic for Automated Machine Learning in Network Systems". With Prof. Busson (U Lyon).
  First job: Postdoc at UC Louvain.
- Youssouph Faye 2022 - 2025
  "Distributed Edge Cloud Architecture for Executing AI-Based Applications".
  First job: Research Engineer at REQUEA.
- Razanne Abu-Aisheh 2020 - 2023
  "Context-Aware Information Gathering and Processing Towards Supporting Autonomous Systems in Industry 4.0 Scenarios". With Dr. Thomas Watteyne (Inria) and Dr. Myriana Rifai (Nokia Bell Labs).
  First job: Research associate at Bristol University.

Postdocs

- Adrien Schoen 2025 - 2026
  "AI for network management". Next position: Postdoc at The University of Hong Kong.
- Francescomaria Faticanti 2023 - 2025
  "Network inference at the edge". Next position: Inria SFP.

Interns

- Matteo Franzil, Luca Giacometti, Greg Depoire–Ferrer, Alain Gérard Mbo, Syed Muhammad Ali Shah, Alyce Théobald, Gaelle Manuela Yonga Yonga, Alexis Carré, Ciro Guida, Jules Marmier, Malo Jaffré, Gaetan Nodet, Duncan Bordin, Hakim Ouhida, Setareh Nouri, Ayoub Ben Ameur.

AWARDS AND HONORS

- Comcast Faculty Innovation Award 2026.
- Junior fellow of the Institut universitaire de France (IUF) 2025-2030.
- Recipient of the GDR RSD Junior Researcher Award 2025.
- Recipient of the George Shultz Innovation Fund ($200k Startup Fund)
- Wall Street Journal Front Page Feature (August 19, 2019, "The Truth About Faster Internet: It's Not Worth It").
- 2016/2017 recipient of the "ECE Graduate Program Academic Achievement Award" from the ECE department at Rutgers University.
- Recipient of the best paper award at the RAFNET 2017 workshop.
- Recipient of the best paper award at the ACM 2012 MobiArch workshop.

PRESS

- "Rethinking Geolocation on the Internet", Internet Society Pulse.
- "Enhancing Internet Resilience with Spatial Analysis", Internet Society Pulse.
- "The Truth About Faster Internet: It's Not Worth It". The Wall Street Journal, Aug 21st 2019 front page print and online editions.
- "SES and Rutgers University test satellite content delivery network for streaming, OTT, and 5G". SES Blog.
- "WINLAB develops infrastructure for potential new internet", The Daily Targum, Nov 2nd 2016

PROFESSIONAL SERVICE

Technical Program Committee Chair: ACM Network Traffic Classification - Reloaded Workshop co-located with ACM CoNEXT 2026 (NTC-R 2026)

Technical Program Committee Member: ACM SIGCOMM 2026, ACM CoNEXT 2026, ACM IMC 2026, ACM Sigmetrics 2026, ACM NewGeo Workshop 2026, ACM HotNets 2025, ACM CoNEXT 2025, ACM Sigmetrics 2025, ACM IMC 2025, USENIX NSDI 2025, IEEE Infocom 2025, ACM IMC 2024, ACM CoNEXT 2024, IEEE Infocom 2024 (Distinguished Member), IEEE MedComNet 2024, CoRes 2024, ACM Sigmetrics 2023 (best paper award committee), IEEE Infocom 2023 (Distinguished Member), CoRes 2023, IEEE MedComNet 2023, IEEE Smartcomp 2023, IEEE CloudCom 2023, IEEE Infocom 2022 (Distinguished Member), IEEE MedComNet 2022, IEEE Smartcomp 2022, IEEE MedComNet 2021, ACM GoodIT 2021, ITC 33, IEEE Infocom 2021, IEEE Infocom 2020, ACM FRUGALTHINGS 2020 Workshop, IEEE Infocom 2019 (Distinguished Member), IEEE VTC2018-Fall, ACM NEAT 2018 Workshop, ACM MMSys 2018, ACM MobiMWareHN 2017 Workshop, ACM S3 2016 Workshop.

Organization Committee Member: IEEE CloudCom 2023, IEEE SMARTCOMP 2020, IEEE CCNC 2018, IEEE CCNC 2017, ACM MobiMWareHN 2017 Workshop, IEEE CCNC 2016, ACM MobiCom 2016, ACM MobiSys 2015.

Editor: Editor for Elsevier Computer Networks (from 2026), Associate Editor for Elsevier Computer Communications (2021-2026).

Reviewer: IEEE TMC, IEEE TSC, ACM TOMM, Elsevier Computer Networks, etc.

Proposal reviewer: Reviewer for grant proposals for the ANR 2024 ASTRID program. Reviewer for grant proposals for the Swiss National Science Foundation (2018).

ACADEMIC SERVICE

Academic duties

Member of the LIP laboratory council. 2025 - Now
Coordinator for the LIP laboratory at of the Carnot de site IMPULSE Lyon Saint-Etienne. 2026 - Now
Coordinator for the Networking Reading Group at ENS Lyon. 2022 - 2026
Correspondent for for the CS department for the green transition at ENS Lyon. 2023 - Now
Head of international relations for the TM department at IAE Savoie Mont Blanc. 2021 - 2022
Animation committee and communication committee at LISTIC Laboratory. 2021 - 2022

Hiring committee member

Maître de conférences in computer science at ENS de Lyon / LIP. 2026
Maître de conférences in computer science at Université Grenoble Alpes / LIG. 2026
CDD Faculty position in computer science at INSA Lyon / CITI. 2026
Maître de conférences in computer science at Sorbonne Université / LIP6. 2025
Enseignants contractuel LRU, at Université d'Avignon / LIA. 2025

Ph.D. jury member

Ph.D. CSI of Gaelle Yonga, Université de Lorraine. 2026-2027
Ph.D. proposal and defense of Taveesh Sharma, University of Chicago. 2026
Ph.D. proposal and defense of Xi Jiang, University of Chicago. 2026
Ph.D. defense of Shurok Khozam, Telecom SudParis. 2026
Ph.D. thesis reviewer and defense of Omar Anser, Inria Nancy. 2026
Ph.D. proposal and defense of Sri Pramodh Rachuri, Stony Brook University. 2026
Ph.D. defense of Chao Wang, EURECOM. 2025
Ph.D. CSI of Gabby Rimlinger, Sorbonne Université. 2024-2025
Ph.D. defense of Davide Aguiari, Università degli Studi di Cagliari and Sorbonne Université. 2021

MAIN REFERENCES

Dipankar Raychaudhuri, Distinguished Professor, Rutgers University, `ray@winlab.rutgers.edu`
Renata Cruz Teixeira, Senior Research Scientist at Netflix and Inria, `rteixeira@netflix.com`
Nick Feamster, Neubauer Professor, University of Chicago, `feamster@uchicago.edu`